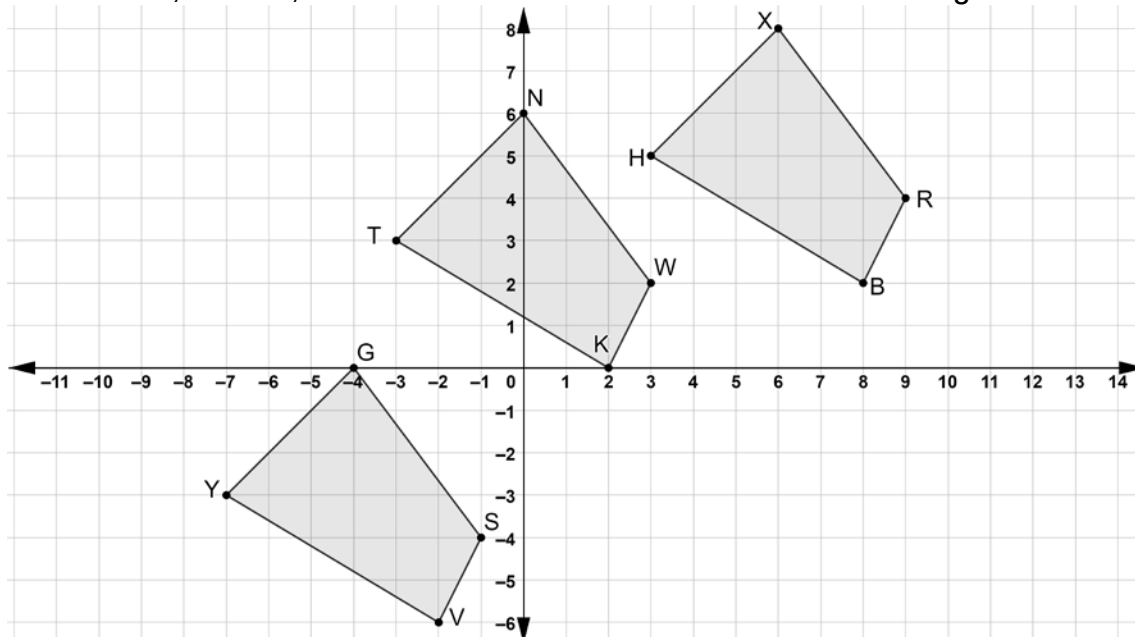


Quadrilaterals $GSVY$, $KWNT$, and $HBRX$ are shown on the coordinate grid.



- Draw the mapping that shows the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.
- Describe the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.
- Complete the statements.
The x -values of quadrilateral $GSVY$, the image, can be found by

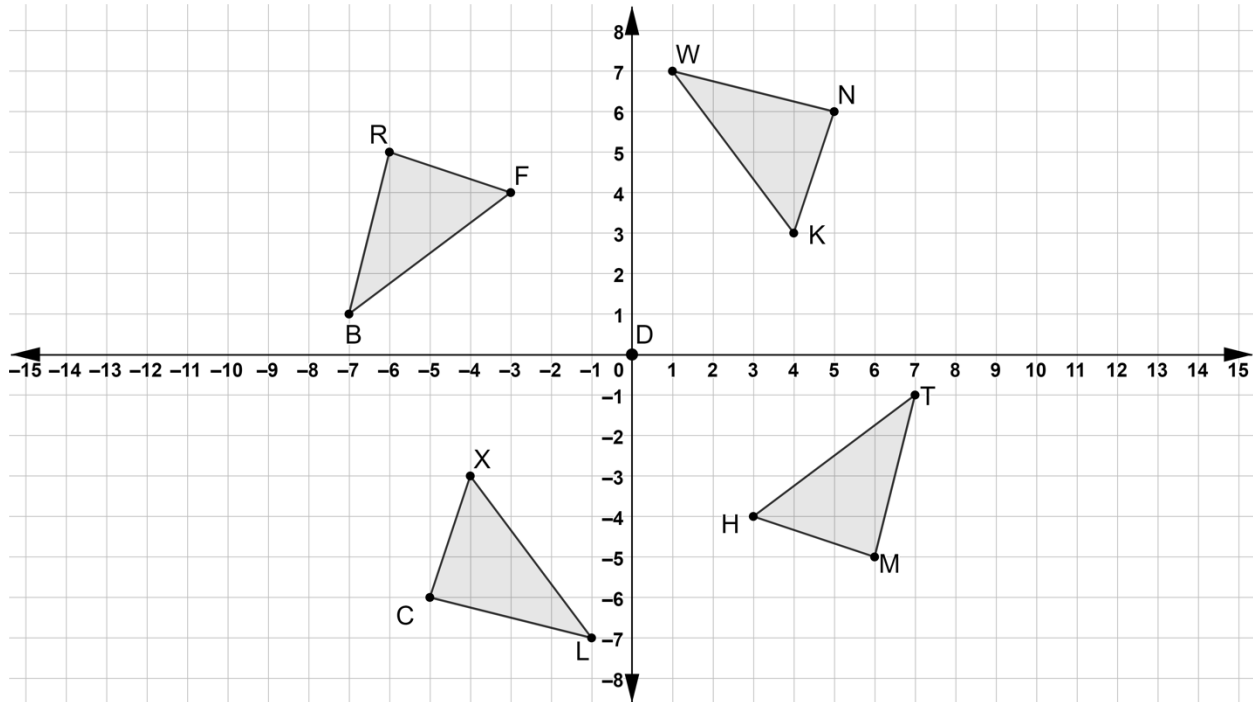
☐ adding
☐ subtracting
☐ multiplying

 _____ to the x -values of quadrilateral $XRHB$, the pre-image. The y -values of quadrilateral $GSVY$, the image, can be found by

☐ adding
☐ subtracting

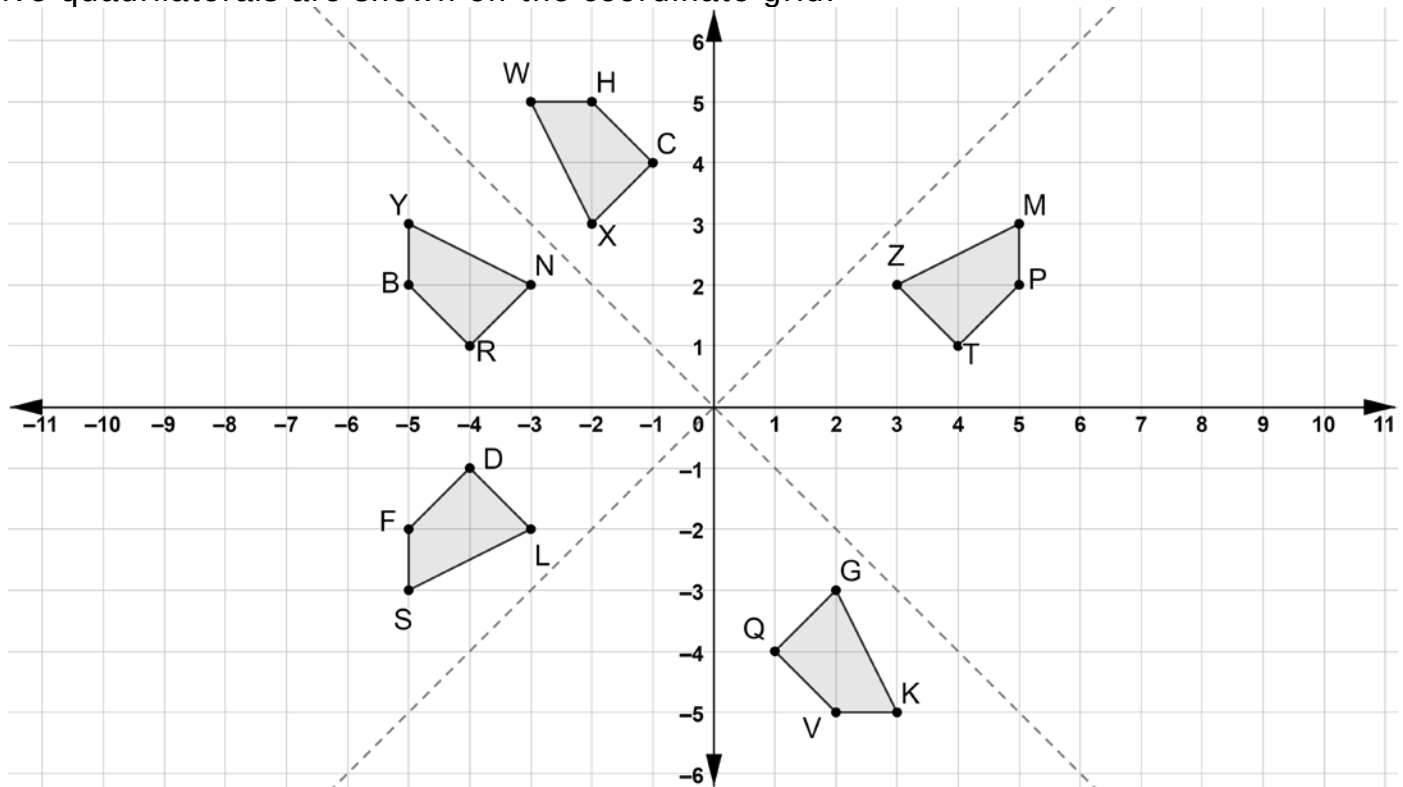
 _____ units to the y -values of quadrilateral $XRHB$, the pre-image.
- Describe the translation if quadrilateral $XRHB$ is the preimage and quadrilateral $GSVY$ is the image.

Four triangles are shown on the coordinate grid. Triangles WNK , TMH , and CXL are all rotations of $\triangle RFB$.



5. On the coordinate grid draw angle FDK .
6. Describe how the measure of $\angle FDK$ can be used to determine the degrees of rotation where $\triangle RFB$ is the pre-image and $\triangle NKW$ is the image.
7. Describe how the coordinates of the vertices of $\triangle RFB$, the preimage, and the coordinates of $\triangle NKW$, the image can be used to determine the degrees of rotation.
8. Describe the rotation where $\triangle RFB$ is the preimage and $\triangle CXL$ is the image.
9. Describe how the coordinates of the vertices of $\triangle RFB$ and $\triangle TMH$ show that rotating $\triangle RFB$ 180° about the origin results in $\triangle MHT$.
10. Explain why the rotation of $\triangle RFB$ that results in $\triangle MHT$ can be described as 180° clockwise or 180° counterclockwise.

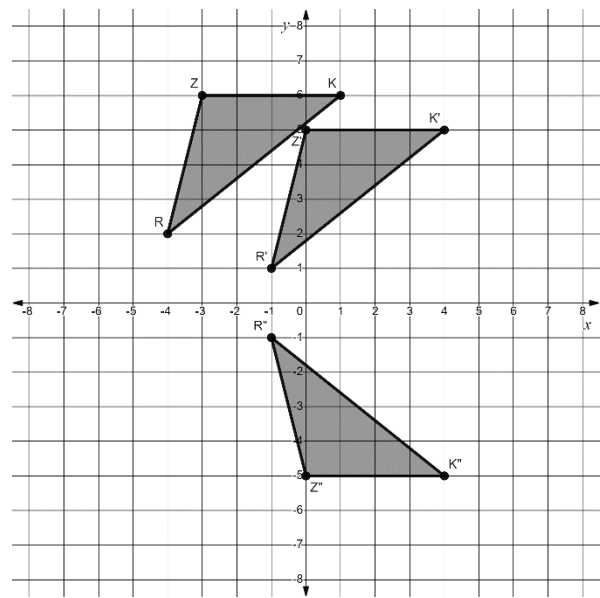
Five quadrilaterals are shown on the coordinate grid.



1. Draw the reflection of quadrilateral $MPTZ$ across the line $y = x$. Label the reflection as $M'P'T'Z'$.
2. Draw the reflection of quadrilateral $VKGQ$ across the line $y = -x$. Label the reflection as $V'K'G'Q'$.
3. Select all of the quadrilaterals that are reflections of quadrilateral $YBRN$.
 - ☐ Quadrilateral $WHCX$
 - ☐ Quadrilateral $MPTZ$
 - ☐ Quadrilateral $KVQG$
 - ☐ Quadrilateral $SFDL$
 - ☐ Quadrilateral $M'P'T'Z'$
 - ☐ Quadrilateral $K'V'Q'G'$
4. The algebraic description of the reflection of quadrilateral $YBRN$, the preimage, onto quadrilateral $PMZT$, the image, is $(x, y) \rightarrow (-x, y)$. Describe how the coordinates of the vertices of quadrilateral $YBRN$, the preimage, and the coordinates of quadrilateral $PMZT$, the image, can be used to determine the line of reflection.

5. Which quadrilateral is the image of reflecting quadrilateral $DLSF$ across the x -axis?
6. Which quadrilateral is the image of reflecting quadrilateral $M'P'T'Z'$ across the y -axis?
7. Describe how the coordinates of the vertices of quadrilateral $BYNR$, the preimage, and the coordinates of quadrilateral $FSLD$, the image, can be used to determine the algebraic description of the reflection.
8. Describe how the coordinates of the vertices of quadrilateral $WHCX$, the preimage, and the coordinates of quadrilateral $YBRN$, the image, can be used to determine the algebraic description of the reflection.
9. Write the algebraic description for the reflection of quadrilateral $BYNR$, the preimage, onto quadrilateral $PMZT$, the image.
10. Write the algebraic description for the reflection of quadrilateral $V'K'G'Q'$, the preimage, onto quadrilateral $HWXC$, the image.

Triangles RZK , $R'Z'K'$, and $R''Z''K''$ are shown on the coordinate grid where $\triangle RZK$ is the preimage.



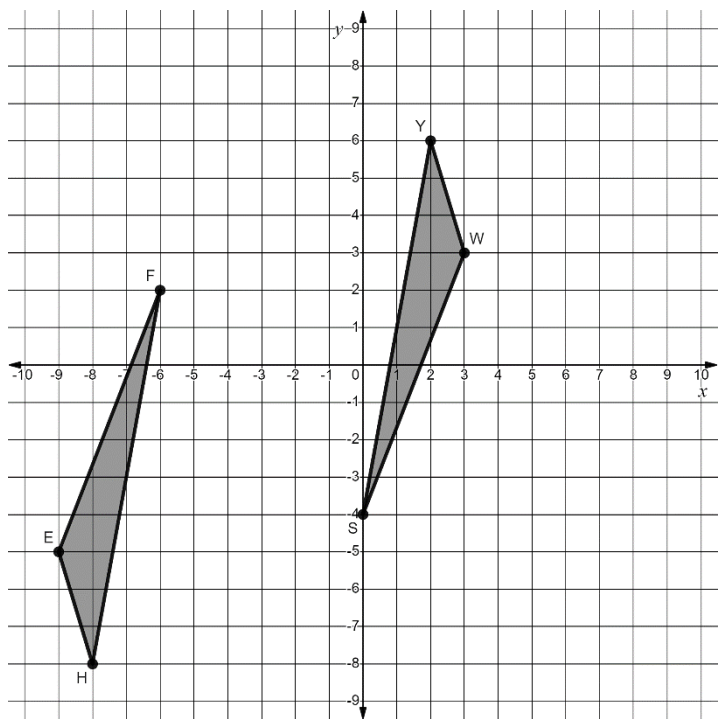
- 1. Describe the feature of $\triangle R'Z'K'$ that indicates the rigid motion that was applied to $\triangle RZK$ that resulted in $\triangle R'Z'K'$.
- 2. Show the feature described by mapping the transformation of one vertex of $\triangle RZK$ to $\triangle R'Z'K'$.
- 3. Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle RZK \rightarrow \triangle R'Z'K'$	

- 4. Map the transformation of one vertex of $\triangle R'Z'K'$ to $\triangle R''Z''K''$.
- 5. Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle R'Z'K' \rightarrow \triangle R''Z''K''$	

Triangles EFH and WSY are shown on the coordinate grid where $\triangle EFH$ is the preimage.

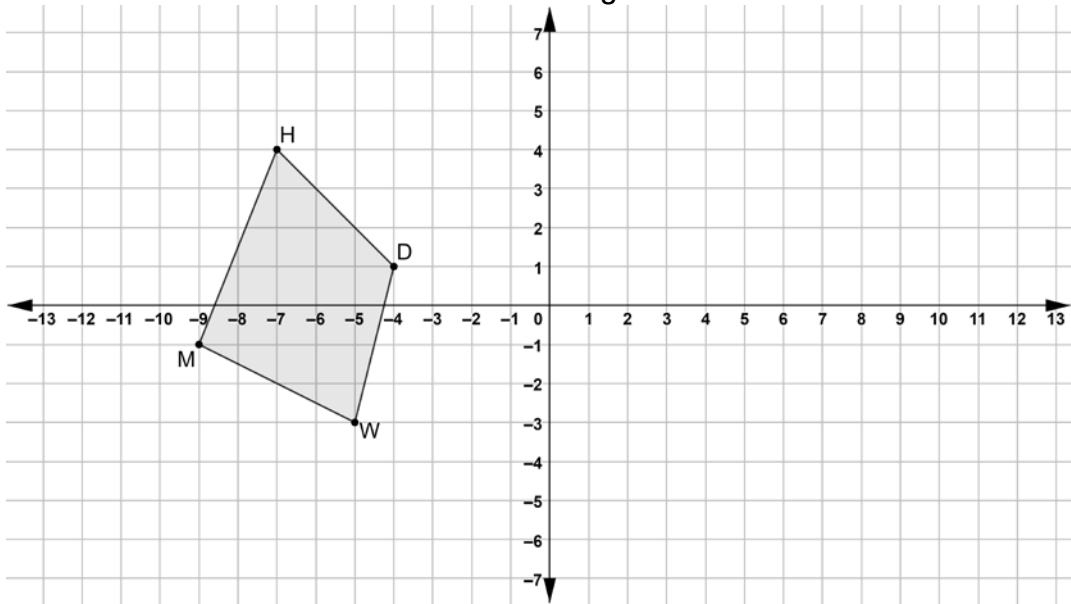


- 6. Describe the feature of $\triangle WSY$ that indicates that more than one rigid motion was applied to $\triangle EFH$.
- 7. Draw $\triangle E'F'H'$, the result of the first transformation that was applied to $\triangle EFH$.
- 8. Map the transformation of one vertex of $\triangle E'F'H'$ to $\triangle WSY$.
- 9. Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle EFH \rightarrow \triangle E'F'H'$	
	$\triangle E'F'H' \rightarrow \triangle WSY$	

- 10. Explain how you know if the sequence of transformations preserves distance.

Quadrilateral $HDWM$ is shown on the coordinate grid.



1. Draw the quadrilateral that is the result of a translation right 8 units and up 3 units and then reflected across the y -axis. Label the new quadrilateral $YBKT$.

Complete the table.

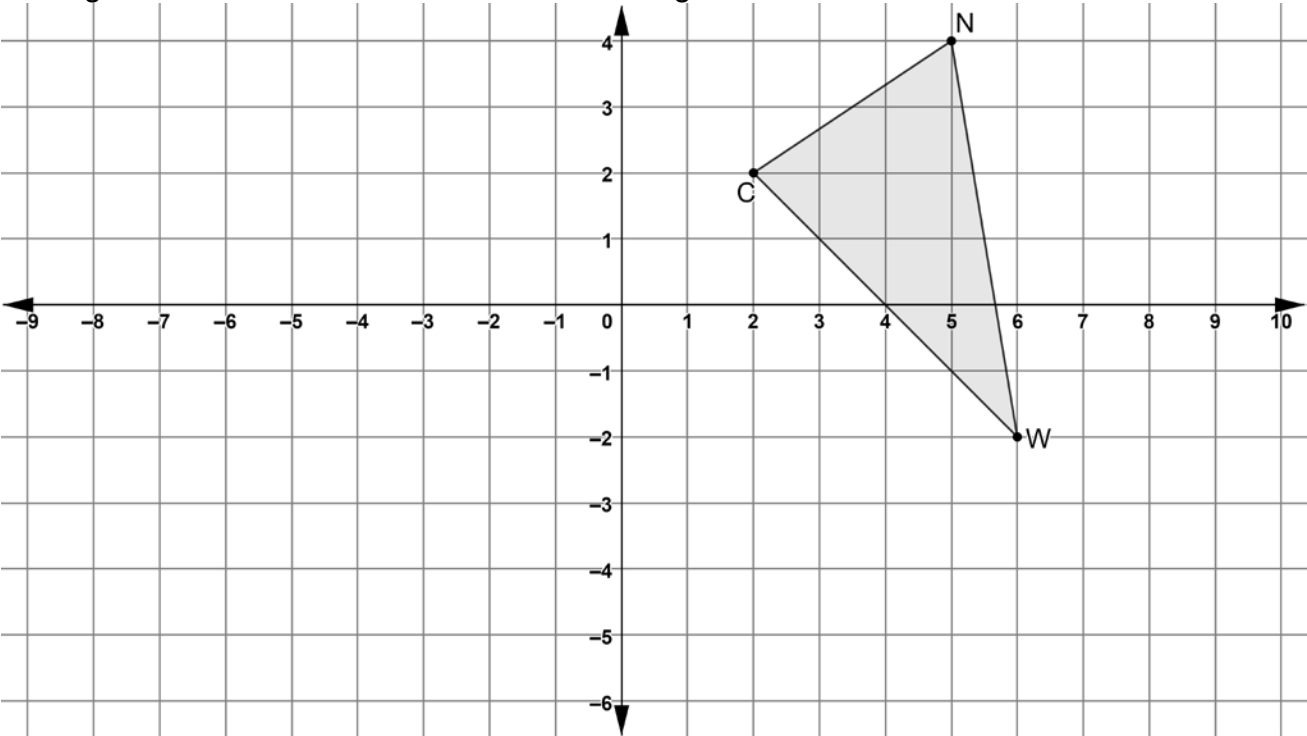
Transformation	Mapping	Algebraic Description
2.	$HDWM \rightarrow H'D'W'M'$	
3.	$H'D'W'M' \rightarrow YBKT$	

4. Complete the table of values for each quadrilateral.

Quadrilateral $HDWM$			Quadrilateral $H'D'W'M'$			Quadrilateral $YBKT$		
Vertex	x	y	Vertex	x	y	Vertex	x	y
H			H'			Y		
D			D'			B		
W			W'			K		
M			M'			T		

5. Explain how the tables for quadrilaterals $HDWM$ and $H'D'W'M'$ show the effect of the transformation on the coordinates that result in the algebraic description.
6. Explain how the tables for quadrilaterals $H'D'W'M'$ and $YBKT$ show the effect of the transformation on the coordinates that result in the algebraic description.

Triangle NWC is shown on the coordinate grid.



7. Draw the triangle that is the result of a rotation 90° clockwise about the origin and then a translation of left 3 units and up 4 units. Label the new triangle $N''W''C''$.

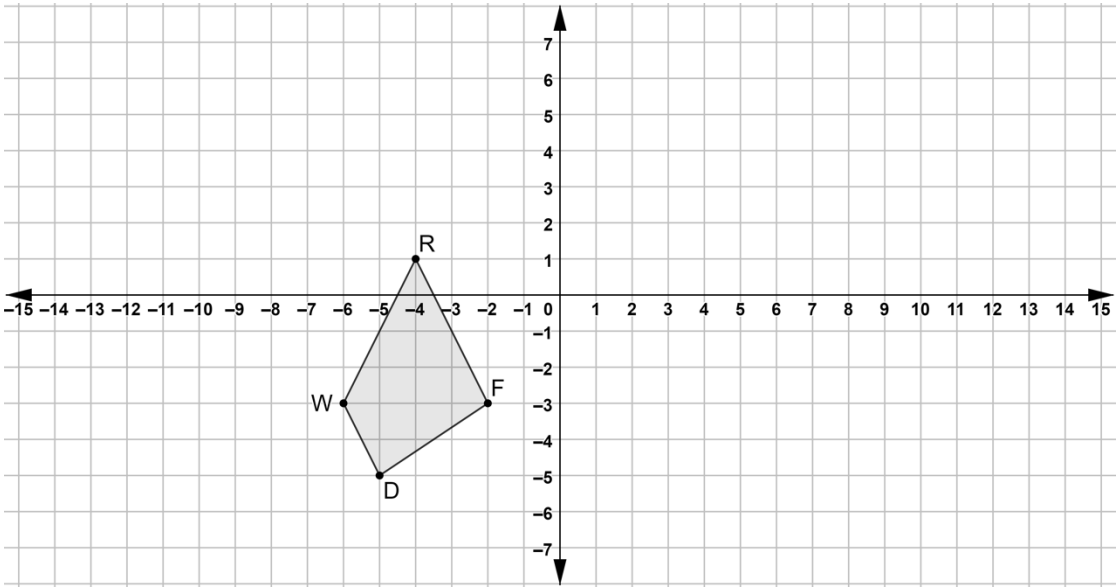
Complete the table.

Transformation	Mapping	Algebraic Description
8.	$\triangle NWC \rightarrow \triangle N'W'C'$	
9.	$\triangle N'W'C' \rightarrow \triangle N''W''C''$	

10. Describe the connection between the algebraic description of the rotation and the vertices of $\triangle NWC$ and $\triangle N'W'C'$.

Quadrilateral $WRFD$ is shown on the coordinate grid.

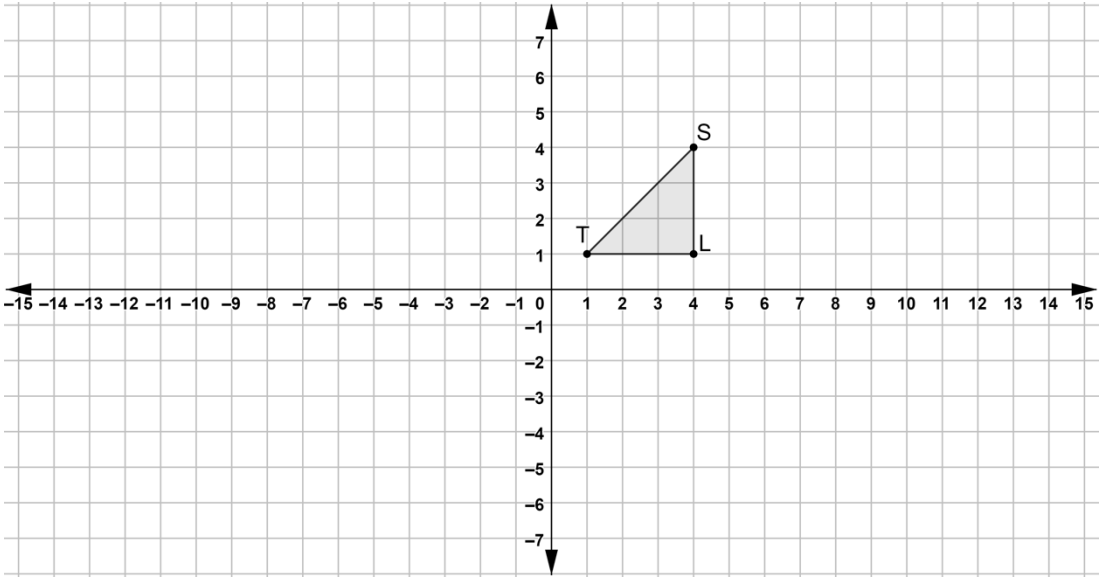
Perform the series of transformations on quadrilateral $WRFD$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
1.	$(x, y) \rightarrow (-y, x)$	$W'R'F'D'$
2.	$(x, y) \rightarrow (x, -y)$	$W''R''F''D''$

Triangle TLS is shown on the coordinate grid.

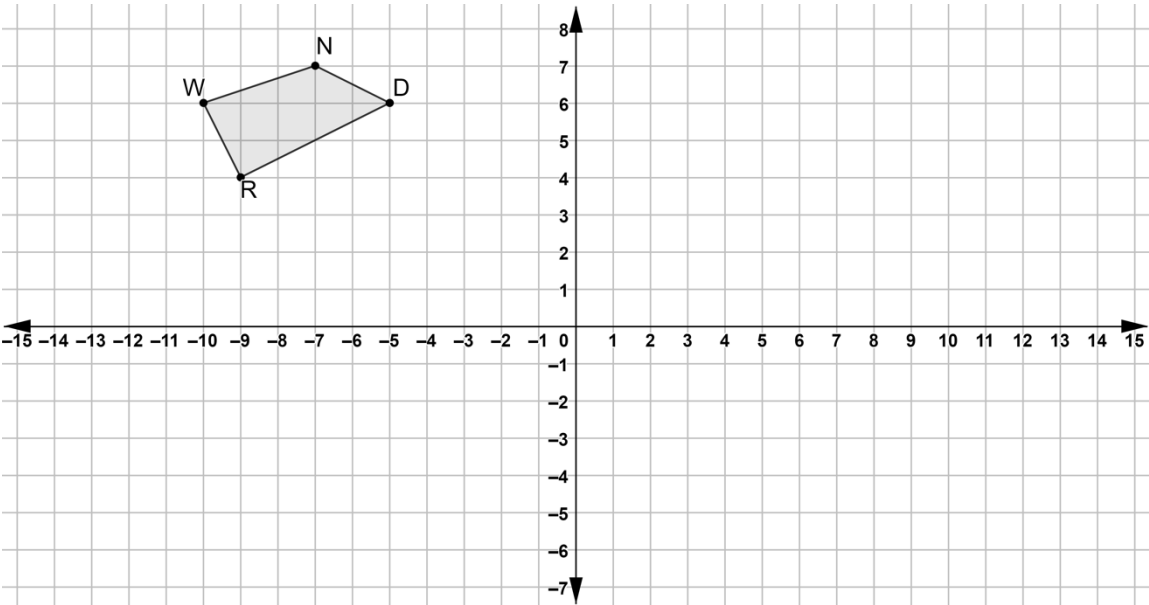
Perform the series of transformations on $\triangle TLS$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Triangle
3.	$(x, y) \rightarrow (-y, x)$	$T'L'S'$
4.	$(x, y) \rightarrow (x - 3, y - 3)$	$T''L''S''$

Quadrilateral $WNDR$ is shown on the coordinate grid.

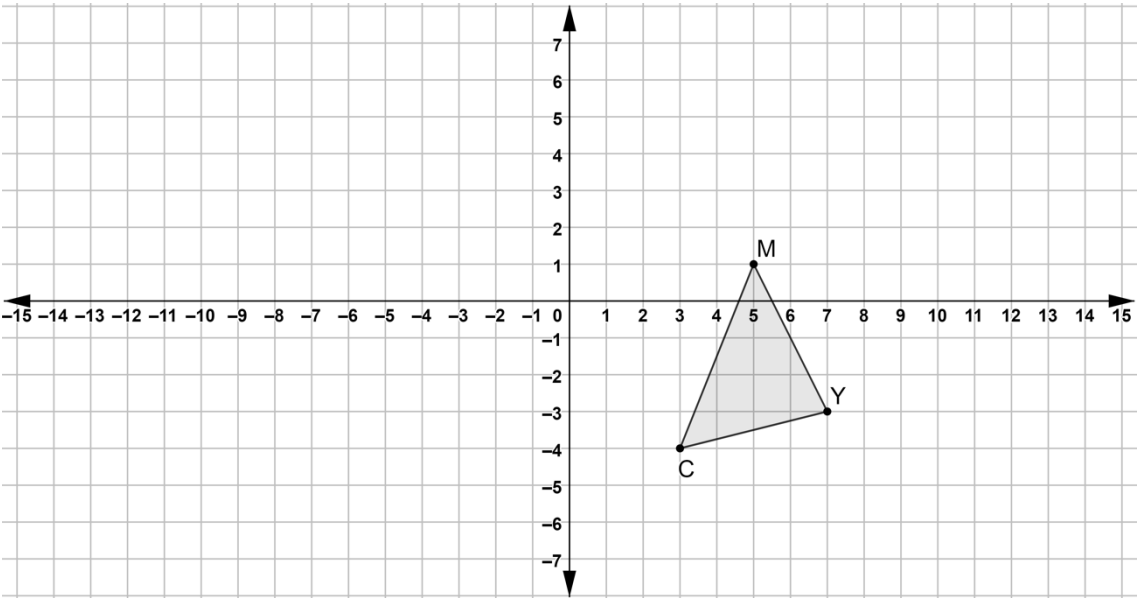
Perform the series of transformations on quadrilateral $WNDR$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
5.	$(x, y) \rightarrow (x + 3, y - 4)$	$W'N'D'R'$
6.	$(x, y) \rightarrow (-x, y)$	$W''N''D''R''$
7.	$(x, y) \rightarrow (-y, x)$	$W'''N'''D'''R'''$

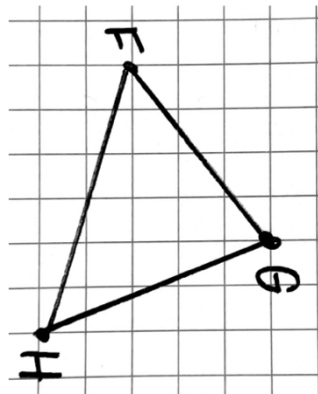
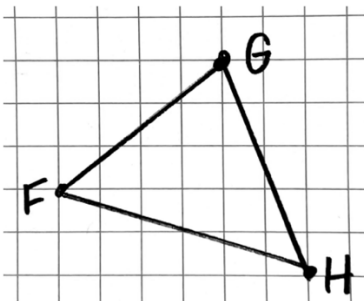
Triangle MYC is shown on the coordinate grid.

Perform the series of transformations on $\triangle MYC$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Triangle
8.	$(x, y) \rightarrow (x - 7, y - 1)$	$M'Y'C'$
9.	$(x, y) \rightarrow (x, -y)$	$M''Y''C''$
10.	$(x, y) \rightarrow (y, -x)$	$M'''Y'''C'''$

Tamika drew $\triangle FGH$ on graph paper. She then rotated the graph paper and moved it to the right.

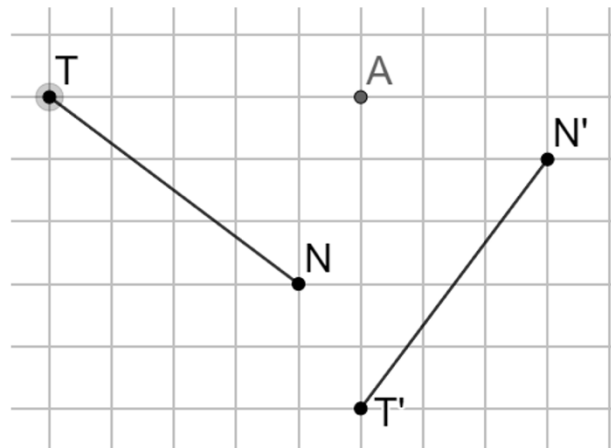


1. Explain what method(s) can be used to determine if each side of $\triangle FGH$ has the same length as the triangle that has been rotated and moved.
2. Explain what method(s) can be used to determine if each angle of $\triangle FGH$ has the same measure as the triangle that has been rotated and moved.
3. Explain why showing that the side lengths and the angles have the same measure shows that the transformations used can be classified as isometry.

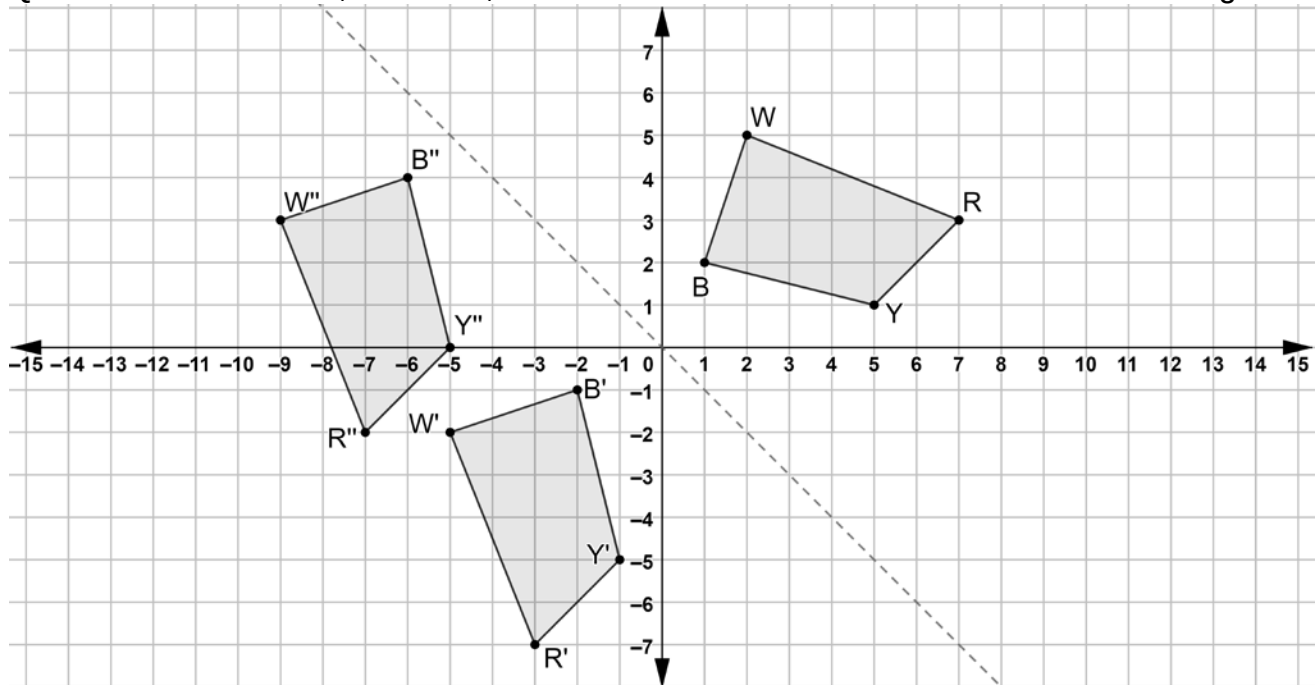
Line segments TN , $T'N'$, and point A are shown on the grid.

4. Draw $\angle TAT'$. Label the angle measure.
5. Describe the transformation applied to $\overline{TN}$ that resulted in $\overline{T'N'}$.
6. Complete the statement.

There is a (n) _____ between the set of points on $\overline{TN}$ and the set of points on $\overline{T'N'}$, therefore all of the points on _____ have to have a matching point on _____.



Quadrilaterals $BWRY$, $B'W'R'Y'$, and $B''W''R''Y''$ are shown on the coordinate grid.

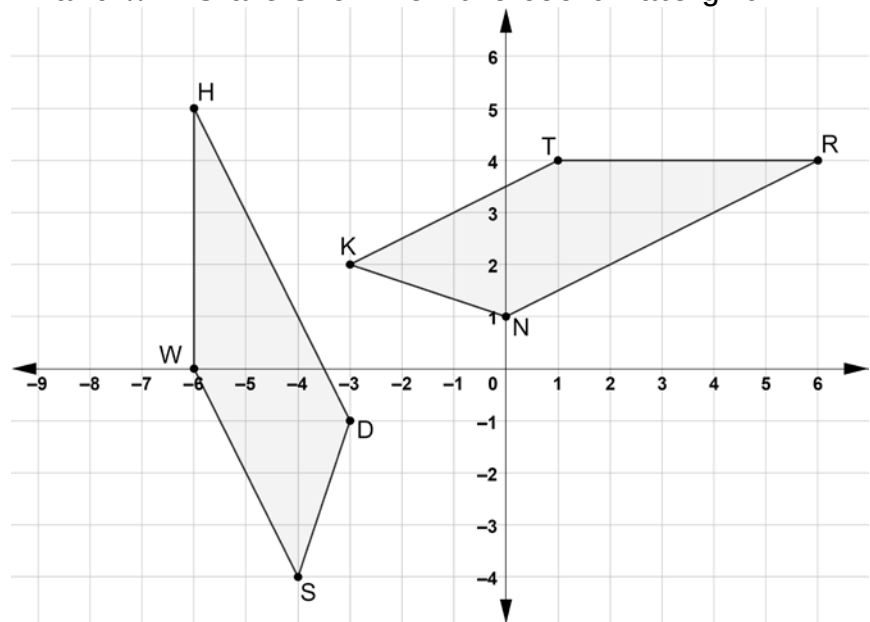


7. Describe what movements should be done with a patty paper copy of quadrilateral $BWRY$ so that the copy of $BWRY$ lays on top of $B'W'R'Y'$.
8. Describe what movements should be done with a patty paper copy of quadrilateral $B'W'R'Y'$ so that the copy of $B'W'R'Y'$ lays on top of $B''W''R''Y''$.
9. Complete the statements.
 The transformations used in both sequences
☐ are
☐ are not

 examples of isometry. All of the points of
☐ $B'W'R'Y'$
☐ $B''W''R''Y''$
☐ $B'W'R'Y'$ and $B''W''R''Y''$

 can be mapped to all of the points of $BWRY$.
10. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $BWRY$ resulted in $BWRY \cong B''W''R''Y''$.

Quadrilaterals $TRNK$ and $WHDS$ are shown on the coordinate grid.



- Describe a sequence of transformations that will map quadrilateral $TRNK$ onto quadrilateral $WHDS$.
- The sequence of transformations that will map quadrilateral $TRNK$ onto quadrilateral $WHDS$ includes a rotation. Describe the sequence using a different rotation.
- Complete the table that shows which vertices were mapped to each other when quadrilateral $TRNK$ was mapped to quadrilateral $WHDS$.

Quadrilateral $TRNK$	Quadrilateral $WHDS$
T	
R	
N	
K	

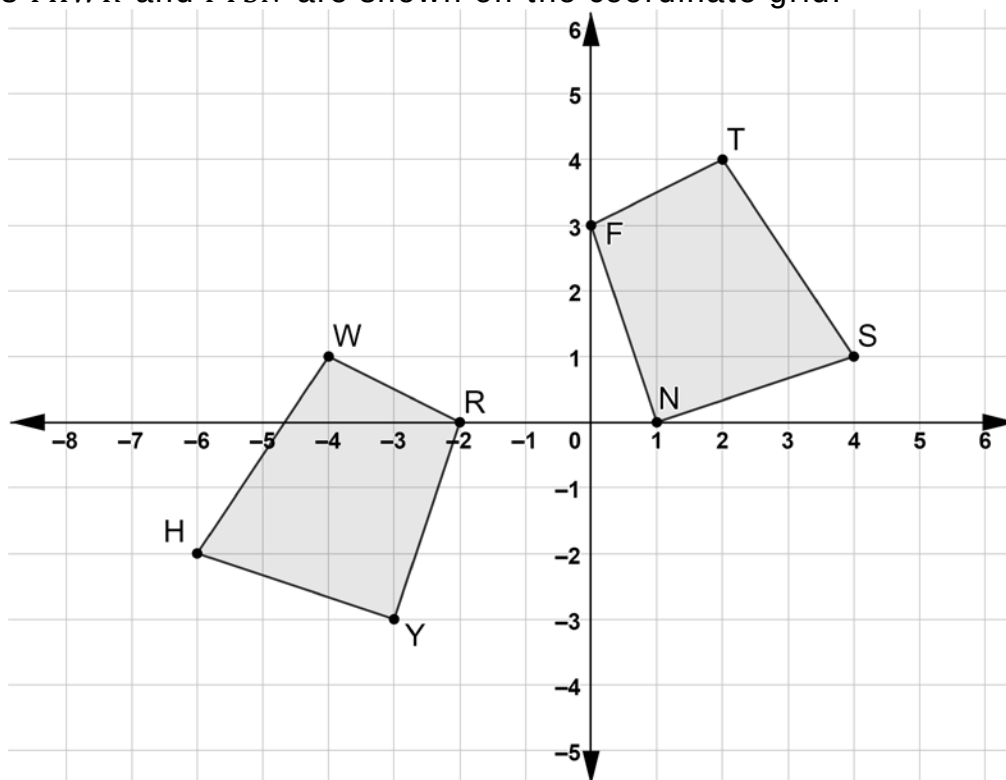
- Write the algebraic description that matches the written description one of the transformations.

$$(x, y) \rightarrow \underline{\hspace{2cm}}$$

$$(x, y) \rightarrow \underline{\hspace{2cm}}$$

- Complete the congruence statement. $TRNK \cong \underline{\hspace{2cm}}$

Quadrilaterals $YHWR$ and $FTSN$ are shown on the coordinate grid.



6. Describe a sequence of transformations that will map quadrilateral $YHWR$ onto quadrilateral $FTSN$.

7. Show the mapping of the transformations on the coordinate grid.

8. Complete the statement.

Vertex Y maps to vertex _____, vertex W maps to vertex _____, vertex H maps to vertex _____, and vertex R maps to vertex _____.

9. Write the algebraic description that matches the written description of the transformations.

$(x, y) \rightarrow$ _____

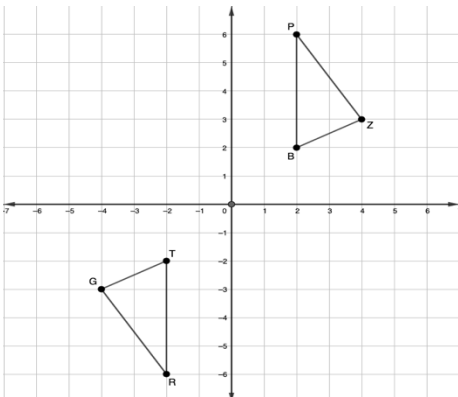
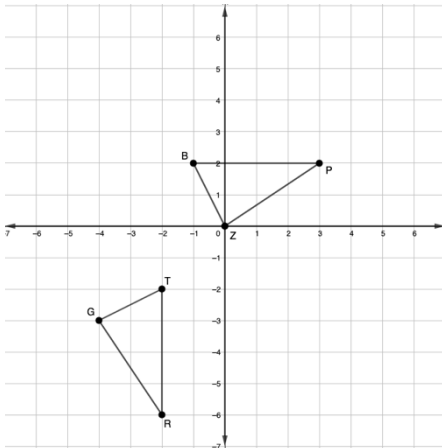
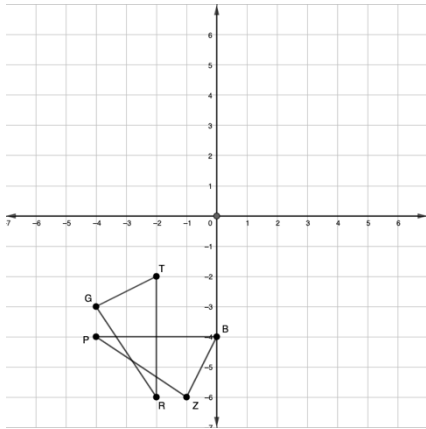
$(x, y) \rightarrow$ _____

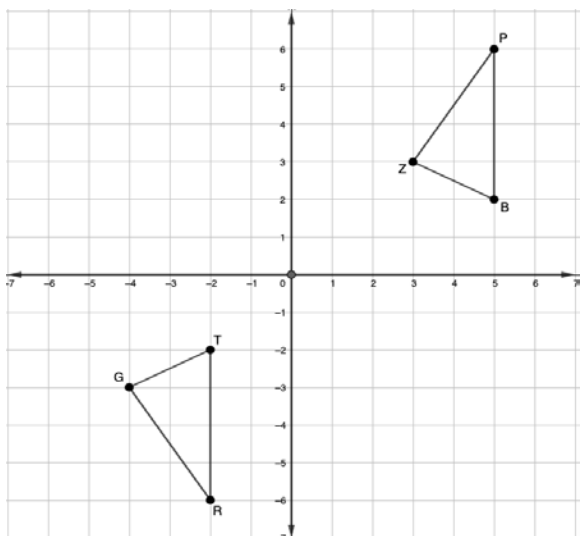
10. Complete the congruence statement. $WRYH \cong$ _____

Geometry
Unit 4
Lesson 8 Homework

Name _____
Date _____
Period _____

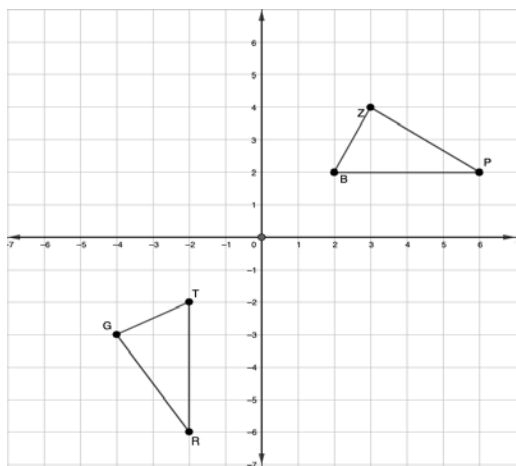
For each coordinate grid with the pre-image of $\triangle TGR$ and image $\triangle BZP$, where $\triangle TGR \cong \triangle BZP$. Complete the table with the missing written and algebraic description(s) that map $\triangle TGR$ onto $\triangle BZP$.

Coordinate Grid	Written Description	Algebraic Description
	1.	2.
	3.	4.
	5.	6.



7.

8.



9.

10.